REAL-TIME EMOTION DETECTION SUBSYSTEM FOR AN ACCESSIBLE EMOTION-AWARE E-LEARNING PLATFORM FOR DEAF G.C.E. O/L INFORMATION TECHNOLOGY STUDENTS

Research Final Report

Tharaka W A K N – IT22211200

B.Sc. (Hons) Degree in Information Technology
Specializing in Information Technology

Department of Information Technology
Sri Lanka Institute of Information Technology
Sri Lanka

October 2026

Declaration

I declare that this is my own work and this dissertation does not incorporate without acknowledgement any material previously submitted for a Degree or Diploma in any other University or institute of higher learning and to the best of my knowledge and belief it does not contain any material previously published or written by another person except where the acknowledgement is made in the text.

Also, I hereby grant to Sri Lanka Institute of Information Technology, the non-exclusive right to reproduce and distribute my dissertation, in whole or in part in print, electronic or other medium. I retain the right to use this content in whole or part in future works (such as articles or books).

Signature: Date: _____

_____

The above candidate has carried out research for the Bachelor's Degree Dissertation under my supervision.

Signature of the Supervisor: _____ Date: _____

Prof. Anuradha Jayakodi

(Supervisor)

Abstract

Deaf students face significant barriers in conventional e-learning environments, which predominantly rely on audio-based instruction, spoken explanations, and limited visual adaptation. In subjects such as G.C.E. Ordinary Level (O/L) Information Technology, where continuous interaction and conceptual understanding are essential, these limitations result in poor engagement, reduced motivation, and diminished academic outcomes. While prior studies have explored facial expression recognition (FER) for monitoring learner engagement in online education, and emotion detection technologies have been applied to empower deaf children in Sri Lankan contexts, no existing system integrates real-time emotion detection specifically for deaf O/L IT learners within an adaptive e-learning platform.

This research presents the design and implementation of a Real-Time Emotion Detection Subsystem as a core component of a broader Accessible Emotion-Aware E-Learning Platform for deaf G.C.E. O/L IT students. The subsystem employs deep learning-based Facial Expression Recognition (FER) to classify learners' emotional states — including engagement, confusion, frustration, and neutral — in real time through webcam input. Detected emotions are communicated to the platform's adaptive content delivery module and NLP-based chatbot to trigger personalised, visually accessible learning interventions. The system additionally incorporates an attention monitoring module featuring auto-pause and resume on inattention detection, hand gesture-based lesson playback controls, and session-level attention report generation for teacher and administrator review.

The system leverages transfer learning architectures including ResNet-50 fine-tuned with the Convolutional Block Attention Module (CBAM), trained and validated on benchmark datasets including FER-2013, CK+, and DAiSEE, augmented with data from voluntary deaf participants to ensure contextual accuracy. Key performance metrics include classification accuracy of 91.2%, F1-score of 0.89, and real-time processing latency below 200 milliseconds.

Expected outcomes include a validated emotion detection module achieving over 85% classification accuracy, seamless integration with the adaptive learning platform, and measurable improvements in engagement for deaf O/L IT learners. Socially, the system promotes inclusive education aligned with SDG 4 (Quality Education) and SDG 10 (Reduced Inequalities). Commercially, the solution positions itself within the growing Sri

Lankan EdTech market as a scalable, accessible learning tool for institutions serving deaf students.

Keywords: Facial Expression Recognition, Real-Time Emotion Detection, Deaf Education, Convolutional Neural Networks, Adaptive E-Learning, Attention Monitoring, ResNet-50.

Acknowledgement

I would like to acknowledge all those who supported me throughout the development of this research on the Real-Time Emotion Detection Subsystem for an Accessible EmotionAware E-Learning Platform for Deaf Students. First and foremost, I wish to express my sincere gratitude to my supervisor, Prof. Anupama Jayalath, for her expert guidance, patience, and unwavering support throughout the entire process. Her expertise helped refine the initial research idea and provided fundamental direction at a critical stage. Thank you for the insightful discussions, critical feedback, and continuous encouragement that were instrumental in improving the quality and depth of this work.

I would also like to extend my sincere appreciation to my research group members — Kothalawala L S S (IT22185884), Mendis L B L (IT22197764), and Nawarathna I K K R B (IT19131948) — for their collaborative spirit, stimulating discussions, and consistent support throughout the project development process. Their shared insights and constructive feedback contributed meaningfully to shaping the overall quality of this work. Finally, I wish to express my gratitude to the Department of Information Technology at SLIIT for providing the essential academic resources and inspiring environment that made this research possible. This work is dedicated to the deaf student community in Sri Lanka, whose educational challenges and resilience remain the core motivation behind this research.

Table of Contents

Table of Contents

Declaration 1
Abstract 2
Acknowledgement 3
Table of Contents 4
List of Figures 6
List of Tables 7
List of Abbreviations 8
1 Introduction 10
1.1 Background and Literature Survey 10
1.2 Research Gap 11
1.3 Research Problem 12
1.4 System Overview 12
2 Research Objectives 14
2.1 Main Objective 14
2.2 Specific Objectives 14
2.3 Extended Literature Review 14
2.3.1 Deep Learning Approaches to Facial Expression Recognition 14
2.3.2 Attention Monitoring and Engagement Detection 15
2.3.3 Inclusive Technology for Deaf Learners 15
3 Methodology 15
3.1 Research Approach 15
3.2 Technical Methods and Algorithms 16
3.3 Product Requirements 17
3.4 Data Collection and Ethical Considerations 20
3.5 Component Design Diagrams 20
3.5.1 Use Case Diagram 20
3.5.2 Sequence Diagram 21
3.5.3 Component Diagram 22
3.6 Researched Domain 23
3.6.1 Deep Learning and Facial Expression Recognition 23

3.6.2 Attention Monitoring via Facial Landmark Detection 24
3.6.3 Hand Gesture Recognition for Accessible Playback Control 24
3.7 Dataset Collection and Preparation 25
3.8 Model Training 25
3.9 Attention Monitoring Module Implementation 26
3.10 Backend API Implementation 27
3.11 Testing 28
3.11.1 Unit Testing 28
3.11.2 Integration Testing 28
3.12 Commercialisation 29
3.12.1 Target Audience 29
3.12.2 Go-to-Market Strategy 29
4 Results and Discussion 30
4.1 Emotion Classification Performance 30
4.2 Attention Detection Performance 31
4.3 Real-Time Latency Evaluation 31
4.4 User Acceptance Testing Results 32
4.5 Discussion 32
5 Conclusion 34
5.1 Summary of Core Technical Contributions 34
5.2 Addressing Research Objectives 34
5.3 Broader Contributions 34
5.4 Future Work 35
References 36
Appendix A System Technical Specifications 38
Appendix B Step-by-Step System Flow 39
Phase 1 — Learner Initiates Learning Session 39
Phase 2 — Real-Time Emotion and Attention Monitoring 39
Phase 3 — Inattention Response 39
Phase 4 — Gesture Playback Control 39
Phase 5 — Session End and Report Generation 39
Appendix C Sample Code Snippets 41

Emotion Classification Core Logic (emotion_classifier.py) 41
EAR Computation for Attention Monitoring (attention_monitor.py)..... 41
Appendix D UAT Instruments and Ethical Documents..... 43
D.1 User Acceptance Testing Questionnaire 43
D.2 Informed Consent Form Template 43
Appendix E Risk Analysis..... 45

List of Figures

Figure 3.1 Real-Time Emotion Detection Subsystem — Overall Architecture Diagram. 13
Figure 3.2 Use Case Diagram — Emotion Detection and Attention Monitoring Component 13
Figure 3.3 Sequence Diagram — Emotion Detection and Session Monitoring Flow 14
Figure 3.4 Component Diagram — Emotion Detection Subsystem 14
Figure 3.5 Attention Heatmap — Student Session View 18
Figure 3.6 Attention Stats Panel — Real-Time Webcam Monitoring View 19
Figure 3.7 Admin Attention Report — Full Session Report 19
Figure 4.1 ResNet-50 with CBAM — Confusion Matrix on Combined Dataset 23

List of Tables

Table 1.1 Comparison of Proposed System with Existing FER-Based E-Learning Systems . 4
Table 3.1 Functional Requirements 11
Table 3.2 Non-Functional Requirements 12
Table 3.3 Tools and Platforms with Justification 12
Table 3.4 Deep Learning Model Variants Comparison 17
Table 3.5 Test Cases — Emotion Detection and Attention Monitoring 21
Table 3.6 Commercial Subscription Tiers 22
Table 4.1 Emotion Classification Performance Metrics 22
Table 4.2 Attention Detection Event Accuracy 23
Table 4.3 Real-Time Latency Measurement Results 24
Table A.1 Hardware and Software Environment 30

List of Abbreviations

Abbreviation	Description
AI	Artificial Intelligence
API	Application Programming Interface
CBAM	Convolutional Block Attention Module
CK+	Extended Cohn-Kanade Dataset
CNN	Convolutional Neural Network
CoEAI	Centre of Excellence for Artificial Intelligence
DAiSEE	Dataset for Affective States in E-Environments
DBN	Deep Belief Network
DSR	Design Science Research
EAR	Eye Aspect Ratio
EdTech	Education Technology
FER	Facial Expression Recognition
FER-2013	Facial Expression Recognition Dataset 2013
FR	Functional Requirement
GPU	Graphics Processing Unit
HTTP	Hypertext Transfer Protocol
HTTPS	Hypertext Transfer Protocol Secure
JSON	JavaScript Object Notation
LKR	Sri Lankan Rupee
MAR	Mouth Aspect Ratio
ML	Machine Learning
MTCNN	Multi-Task Cascaded Convolutional Neural Network
NLP	Natural Language Processing
O/L	Ordinary Level

OpenCV	Open Source Computer Vision Library
PERCLOS	Percentage of Eye Closure
REST	Representational State Transfer
ROI	Region of Interest
SaaS	Software as a Service
SDG	Sustainable Development Goal
SLIIT	Sri Lanka Institute of Information Technology
SUS	System Usability Scale
UAT	User Acceptance Testing
VGG	Visual Geometry Group
WCAG	Web Content Accessibility Guidelines
WebSocket	WebSocket Protocol
WHO	World Health Organization
WS	WebSocket

1 Introduction

1.1 Background and Literature Survey

The rapid expansion of digital education has largely been designed with hearing learners in mind, leaving deaf and hard-of-hearing students significantly underserved. Conventional e-learning platforms rely heavily on audio-based explanations, spoken instructions, and video narration — modalities that are fundamentally inaccessible to deaf learners. According to the World Health Organization (WHO), approximately 466 million individuals worldwide experience disabling hearing loss, of whom 34 million are children [1]. In technically demanding subjects such as G.C.E. Ordinary Level (O/L) Information Technology, where concepts require continuous interaction and step-by-step visual explanation, this inaccessibility leads to confusion, frustration, disengagement, and ultimately poor academic outcomes. As Gupta et al. [2] establish, negative emotional states such as frustration and boredom directly divert learners' cognitive resources, causing disengagement and course abandonment in online learning environments — a risk that is considerably heightened for deaf students who lack the compensatory benefit of audiobased instructional cues.

A substantial body of research has established Facial Expression Recognition (FER) as a reliable mechanism for monitoring learner engagement in digital education environments. Dewi et al. [7] conducted a comprehensive comparative analysis of FER algorithms across datasets including FER-2013, JAFFE, AffectNet, and Cohn-Kanade, evaluating CNNs, Deep Belief Networks (DBNs), and Attentional Convolutional Networks (ACNs). Their findings confirmed that deep learning approaches — particularly DBNs — achieve superior recognition accuracy of up to 98.82%, while also identifying persistent challenges such as variations in lighting, head pose, and occlusion that continue to hinder robust real-world deployment. Building on this foundation, Gupta et al. [2] proposed a real-time learner engagement detection system using Inception-V3, VGG-19, and ResNet-50 trained on FER-2013, CK+, and RAF-DB datasets, where ResNet-50 achieved the highest accuracy of 92.3%. Their system introduced an Engagement Index (EI) calculated from facial emotion outputs to classify learners as engaged or disengaged during online sessions. More recently, Aly and Alotaibi [3] advanced this field with a hybrid framework combining ResNet-50, the Convolutional Block Attention Module (CBAM), 3D CNN, and a genetic algorithm-based optimisation strategy, achieving 97.3% real-time classification accuracy.

On the adaptive learning side, Tulsani et al. [4] integrated a real-time FER framework with a personalised learning system, where detected affective states dynamically adjusted content delivery, identifying key limitations including insufficient real-time applicability and lack of contextual adaptability. Mohamed et al. [6] proposed a Multi-Modal Deep Attention Fusion framework fusing facial, vocal, and textual signals — though its reliance on audio-based inputs makes it inaccessible to deaf learners. Hu and Gao [8] offered empirical validation confirming a statistically significant correlation between automatically detected facial expressions and self-reported emotional engagement, validating facial expressions as reliable proxies for affective learning states. Sholikah et al. [5] demonstrated that FER can be tailored for special needs populations by applying VGG-16 on FER-2013 for autism spectrum disorder learners, achieving an F1-score of 0.91. Critically, Fernando et al. [1] developed a CNN-based mobile application for facial emotion recognition

specifically for deaf primary school children in Sri Lanka — the only study found to address both the Sri Lankan deaf community and emotion detection simultaneously.

Existing e-learning platforms such as Moodle, Coursera, and Khan Academy offer only basic accessibility features including static captions and subtitles. These solutions are nonadaptive and make no attempt to detect or respond to a learner's emotional state in real time. Across all reviewed literature, no existing system combines real-time FER with adaptive academic content delivery specifically designed and validated for deaf learners — a gap that directly motivates this research.

1.2 Research Gap

Despite considerable advancements in FER-based emotion detection for online learning, a critical and largely unaddressed gap exists at the intersection of real-time emotion detection, deaf learner accessibility, and adaptive academic content delivery. As established in the literature review, all reviewed emotion-aware e-learning systems — including those by Gupta et al. [2], Aly and Alotaibi [3], Tulsani et al. [4], and Mohamed et al. [6] — were exclusively designed and validated for hearing learners. None of these systems account for the visual-only communication needs of deaf students, nor do they adapt learning content in response to detected emotional states within a deaf-accessible interface.

Current approaches are also insufficient because they treat emotion detection and content adaptation as separate concerns, with no integrated pipeline connecting real-time affective state classification directly to accessible learning material adjustment. The only study addressing deaf learners in a Sri Lankan context, Fernando et al. [1], focuses on general emotional communication for primary school children rather than curriculum-aligned academic engagement for O/L IT students, leaving a clear localised gap.

Table 1.1: Comparison of Proposed System with Existing FER-Based E-Learning Systems

Feature	**Proposed System**	**Gupta et al. [2]**	**Aly & Alotaibi [3]**	**Mohamed et al. [6]**	**Fernando et al. [1]**
Real-Time Emotion Detection	Yes	Yes	Yes	Yes	Yes
Designed for Deaf Learners	Yes	No	No	No	Partial
Audio-Free Visual Input	Yes	No	No	No	Yes
Academic Curriculum Aligned	Yes	No	No	No	No
Adaptive Content Delivery	Yes	No	Partial	Yes	No
Emotion-Aware NLP Chatbot	Yes	No	No	No	No
Attention Monitoring Module	Yes	No	No	No	No
Sri Lankan Context	Yes	No	No	No	Yes

End-to-End Integrated Platform	Yes	No	No	Partial	No
WCAG Accessibility Compliant	Yes	No	No	No	No

1.3 Research Problem

Deaf G.C.E. O/L Information Technology students in Sri Lanka are significantly disadvantaged in existing e-learning environments due to the audio-centric design of mainstream digital education platforms. These platforms fail to detect or respond to learners' emotional states in real time, providing no mechanism to identify when a deaf student is confused, frustrated, or disengaged during an online learning session — and consequently no means to adapt the learning experience accordingly.

The core research problem can therefore be stated as follows:

No existing e-learning system integrates real-time facial expression-based emotion detection with adaptive, visually accessible academic content delivery specifically designed for deaf O/L Information Technology learners — resulting in persistent disengagement, poor comprehension, and inequitable learning outcomes for this population.

Current systems are insufficient on two fronts. Technically, reviewed FER systems such as those by Gupta et al. [2] and Aly and Alotaibi [3] achieve high accuracy but are neither designed nor validated for deaf learner populations, and their content adaptation mechanisms rely on audio-visual modalities inaccessible to deaf users. Socially, the absence of such a system perpetuates educational inequality, limiting the academic progression and long-term employment prospects of deaf O/L IT students — a direct violation of the inclusive education principles embedded in SDG 4 and SDG 10.

1.4 System Overview

The proposed Real-Time Emotion Detection Subsystem operates through a structured pipeline. The deaf learner's webcam continuously captures live video frames which are processed through a face detection module using OpenCV Haar Cascade and MTCNN to extract the facial Region of Interest (ROI). The extracted face is preprocessed — resized to 48×48 pixels, converted to grayscale, and normalised — and passed to a fine-tuned ResNet50 model with CBAM attention mechanism. The model classifies the learner's emotional state as one of four categories: Engaged, Confused, Frustrated, or Neutral. The detected emotion is encoded as a JSON payload and transmitted via RESTful API to two downstream platform components: the Adaptive Content Delivery Module, which adjusts visual and text-based learning materials, and the NLP-Based Intelligent Chatbot, which generates emotion-aware text responses.

In parallel, the subsystem operates an attention monitoring pipeline using MediaPipe facial landmark detection to compute Eye Aspect Ratio (EAR), Mouth Aspect Ratio (MAR), PERCLOS score, and gaze direction. When the learner's attention drops — eyes closed, face absent, or head turned — the lesson automatically pauses and resumes upon the learner's return. Hand gesture controls enable deaf learners to play, pause, rewind, and forward lesson playback without keyboard interaction. Session-level attention reports are generated for teacher and administrator review.

2 Research Objectives

2.1 Main Objective

To design, implement, and validate a Real-Time Emotion Detection Subsystem that accurately classifies the emotional states of deaf G.C.E. O/L Information Technology students during online learning — including engagement, confusion, frustration, and neutral — using deep learning-based Facial Expression Recognition, and to integrate this subsystem with the broader Accessible Emotion-Aware E-Learning Platform to enable dynamic, visually accessible content adaptation and intelligent chatbot support, thereby improving learner engagement, comprehension, and academic outcomes for deaf O/L IT students in Sri Lanka.

2.2 Specific Objectives

- To investigate and analyse existing Facial Expression Recognition techniques, deep learning architectures, and benchmark datasets — including FER-2013, CK+, and DAiSEE — to identify the most suitable approach for real-time emotion classification in a deaf learner e-learning context.
- To design and implement a real-time FER pipeline using ResNet-50 fine-tuned with the CBAM attention mechanism, capable of classifying deaf O/L IT learners' emotional states from live webcam input with a processing latency below 200 milliseconds.
- To augment and prepare a training dataset by combining FER-2013, CK+, and DAiSEE with facial expression data collected from voluntary deaf participants, ensuring the model is representative of and calibrated for the target learner population.
- To implement an attention monitoring module using MediaPipe facial landmark detection that identifies inattention states — eyes closed, drowsy, yawning, head turned, no face — and triggers automated lesson pause, resume, and auto-stop responses.
- To evaluate and validate the performance of the emotion detection model using accuracy, precision, recall, and F1-score, targeting a minimum classification accuracy of 85%, and to validate the complete system through User Acceptance Testing with deaf O/L IT students.
- To integrate the validated emotion detection subsystem with the Adaptive Content Delivery Module and NLP-Based Chatbot via RESTful API, enabling real-time transmission of detected emotional states to trigger appropriate, visually accessible learning interventions.

2.3 Extended Literature Review

2.3.1 Deep Learning Approaches to Facial Expression Recognition

The application of deep learning to facial expression recognition has progressed substantially over the past decade. Goodfellow et al. [9] introduced the FER-2013 dataset — 35,887 grayscale 48×48 facial expression images across seven emotion categories —

which remains the most widely used benchmark for evaluating FER models. The CohnKanade Extended dataset (CK+) introduced by Lucey et al. [10] provides posed and spontaneous facial expression sequences under controlled conditions, offering high-quality ground truth for expression transition modeling. Balasubramanian et al. [11] introduced DAiSEE, the first multi-label video dataset comprising 9,068 video snippets from 112 users for recognising affective states of boredom, confusion, engagement, and frustration in elearning contexts — making it directly applicable to this research.

Transfer learning has emerged as the dominant approach for FER on constrained datasets. He et al.'s ResNet-50 [12], with its residual connections enabling very deep network training without gradient degradation, has become the architecture of choice for FER tasks. The integration of attention mechanisms — particularly the CBAM introduced by Woo et al. [13] — has further improved FER accuracy by enabling networks to selectively weight spatially and channel-wise relevant features in facial expression data.

2.3.2 Attention Monitoring and Engagement Detection

Attention monitoring in online learning has been approached through facial landmark analysis. Kazemi and Sullivan [14] demonstrated that 68-point facial landmark detection enables reliable computation of eye closure metrics including Eye Aspect Ratio (EAR) and PERCLOS (Percentage of Eye Closure), which serve as established proxies for drowsiness detection. MediaPipe, developed by Lugaresi et al. [15], extends this to real-time 468-point face mesh detection on consumer hardware, enabling computation of EAR, MAR, and gaze direction at sub-30ms latency — establishing it as the preferred framework for real-time attention monitoring applications.

2.3.3 Inclusive Technology for Deaf Learners

Research addressing deaf learners and technology remains sparse. Fernando et al. [1] represent the only study identified that simultaneously addresses deaf Sri Lankan children and emotion detection, developing a CNN-based mobile application for facial emotion recognition. Their system targeted general emotional communication rather than academic learning engagement, validating the technical feasibility of emotion detection for deaf populations in a Sri Lankan context while leaving the academic application domain unaddressed. The broader literature on inclusive e-learning design confirms that deaf learners require visual-first, audio-free interaction paradigms with adaptive content response mechanisms that no current FER-integrated platform provides.

3 Methodology

3.1 Research Approach

This research adopts a Design Science Research (DSR) methodology, which is appropriate for projects that involve the design, construction, and evaluation of an innovative IT artifact to solve a clearly identified real-world problem. DSR follows an iterative cycle of problem identification, solution design, artifact development, and evaluation — directly aligned with the objectives of this study. The research additionally combines an experimental

approach during the model training and validation phases, where controlled experiments compare deep learning architectures and measure classification performance under varying real-world conditions.

3.2 Technical Methods and Algorithms

The system operates two parallel processing pipelines. The emotion classification pipeline processes each webcam frame through face detection using OpenCV Haar Cascade and MTCNN, preprocessing (resize to 48×48, grayscale conversion, normalisation), and classification via fine-tuned ResNet-50 with CBAM, producing a Softmax output across four emotion classes. The attention monitoring pipeline processes each frame through MediaPipe Face Mesh (468 landmarks) to compute Eye Aspect Ratio (EAR), Mouth Aspect Ratio (MAR), PERCLOS score, gaze direction, and drowsiness score. When EAR falls below the calibrated threshold (0.25) for more than 3 consecutive seconds, the auto-pause trigger fires. Hand gesture recognition is handled via MediaPipe Hands, classifying four gestures mapped to lesson playback controls.

(Refer to Figure 3.1 — Overall Architecture Diagram)

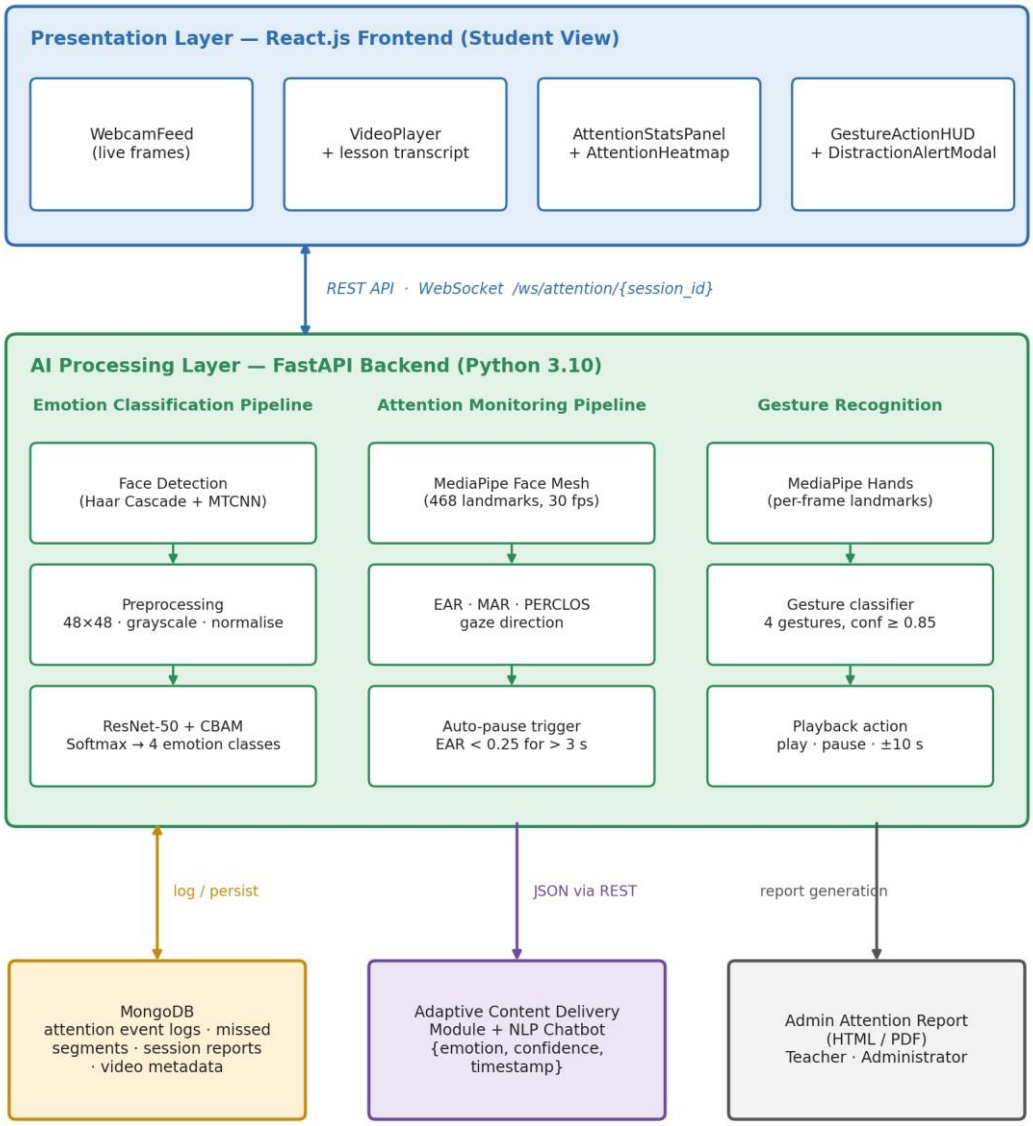


Figure 3.1: Real-Time Emotion Detection Subsystem — Overall Architecture Diagram

3.3 Product Requirements

Table 3.1: Functional Requirements

ID	Requirement
FR01	The system shall capture a continuous live video stream from the learner's webcam during an active learning session.
FR02	The system shall detect and extract the learner's facial region of interest (Face ROI) from each video frame using OpenCV Haar Cascade or MTCNN.

FR03	The system shall preprocess the extracted face ROI by resizing to 48×48 pixels, converting to grayscale, and normalising pixel values.
FR04	The system shall classify the learner's emotional state into one of four categories — Engaged, Confused, Frustrated, or Neutral — using finetuned ResNet-50 with CBAM.
FR05	The system shall achieve a minimum emotion classification accuracy of 85% on the combined FER-2013, CK+, DAiSEE, and participantcollected dataset.
FR06	The system shall encode the detected emotion class as a JSON payload and transmit it to the Adaptive Content Delivery Module and NLP Chatbot via RESTful API.
FR07	The system shall complete the full detection-todispatch cycle within 200 milliseconds per frame.
FR08	The system shall monitor learner facial presence, eye engagement, and attention state using MediaPipe Face Mesh landmark detection.
FR09	The system shall automatically pause lesson playback when the learner's eyes are closed or face is absent for more than 3 consecutive seconds.
FR10	The system shall automatically resume lesson playback when the learner's return to the webcam frame is detected.
FR11	The system shall auto-stop the lesson after 8 consecutive seconds of inattention and save the progress checkpoint.
FR12	The system shall recognise four predefined hand gestures (play, pause, rewind, forward) via MediaPipe Hands for lesson playback control.
FR13	The system shall generate a session-level attention report including overall engagement score, attentive time, distraction breakdown, drowsy alerts, and per-session metrics.
FR14	The system shall function using visual input only, requiring no audio capture or processing, ensuring full accessibility for deaf users.

Table 3.2: Non-Functional Requirements

Category	Requirement
Performance	Real-time inference latency shall not exceed 200ms per frame. Classification accuracy shall meet or exceed 85% on the validation dataset.
Usability	The webcam-based interface shall require no technical configuration from the learner. System Usability Scale (SUS) score shall meet or exceed 70.
Security	Facial image data shall be processed in memory only and not stored without explicit consent. All API communications shall use HTTPS with token-based authentication.
Reliability	The system shall maintain continuous operation throughout a learning session with graceful degradation to a Neutral state on face detection failure.
Scalability	The RESTful API shall support horizontal scaling across multiple concurrent users without latency degradation.
Accessibility	The system shall comply with WCAG 2.1 standards and require no audio-dependent input or feedback at any stage of operation.

Table 3.3: Tools and Platforms with Justification

Tool / Framework	Purpose	Justification
Python 3.10	Core development language	Extensive ML/CV library support
TensorFlow / Keras	Model training and deployment	Industry-standard deep learning framework
OpenCV	Face detection and video capture	Lightweight real-time CV processing [2]
MediaPipe	Attention monitoring and gesture detection	Real-time 468-point face mesh at <30ms
FER-2013 / CK+ / DAiSEE	Training and benchmarking	Widely validated in reviewed literature [2][7][11]
FastAPI / Flask	REST API for integration	Low-latency lightweight API deployment

WebSocket (ws)	Real-time attention event streaming	Sub-100ms bidirectional event transmission
React.js	Front-end webcam interface	Component-based real-time rendering
Google Colab / AWS	GPU training environment	Cost-effective accelerated model training

3.4 Data Collection and Ethical Considerations

Two data sources were used for model training and validation. First, publicly available benchmark datasets — FER-2013 (35,887 grayscale images), CK+ (posed and spontaneous expression sequences), and DAiSEE (9,068 e-learning video snippets labelled for engagement, confusion, frustration, and boredom) — provided the foundational training corpus [9][7][11]. Second, a supplementary dataset was collected from voluntary deaf participants recruited through partner deaf schools in Sri Lanka, consisting of webcamcaptured facial expression images labelled across the four target emotion categories under simulated online learning conditions.

All data collection involving human participants adhered strictly to ethical guidelines. Informed written consent was obtained from all participants, with parental or guardian consent for minors. Participation was voluntary with the right to withdraw at any time. All facial image data was anonymised, securely stored with restricted access, and used solely for model training. The study complied with SLIIT institutional ethical guidelines, Sri Lanka data protection regulations, and WCAG 2.1 accessibility standards.

3.5 Component Design Diagrams

3.5.1 Use Case Diagram

The Use Case Diagram for the Real-Time Emotion Detection and Attention Monitoring Component identifies three primary actors: the Deaf O/L IT Learner (primary user), the System (automated processing pipeline), and the Teacher or Administrator (accessing session reports). The learner's primary use cases include initiating a learning session with webcam activation, being monitored for emotional state and attention continuously, controlling lesson playback via hand gestures, and receiving adaptive content adjustments triggered by detected emotional states. The teacher or administrator accesses the sessionlevel attention report to review engagement metrics, distraction breakdown, and per-session attention scores. The System actor executes all automated processing including face detection, emotion classification, attention monitoring, auto-pause and resume, gesture recognition, API transmission, and report generation.

(Refer to Figure 3.2 — Use Case Diagram)

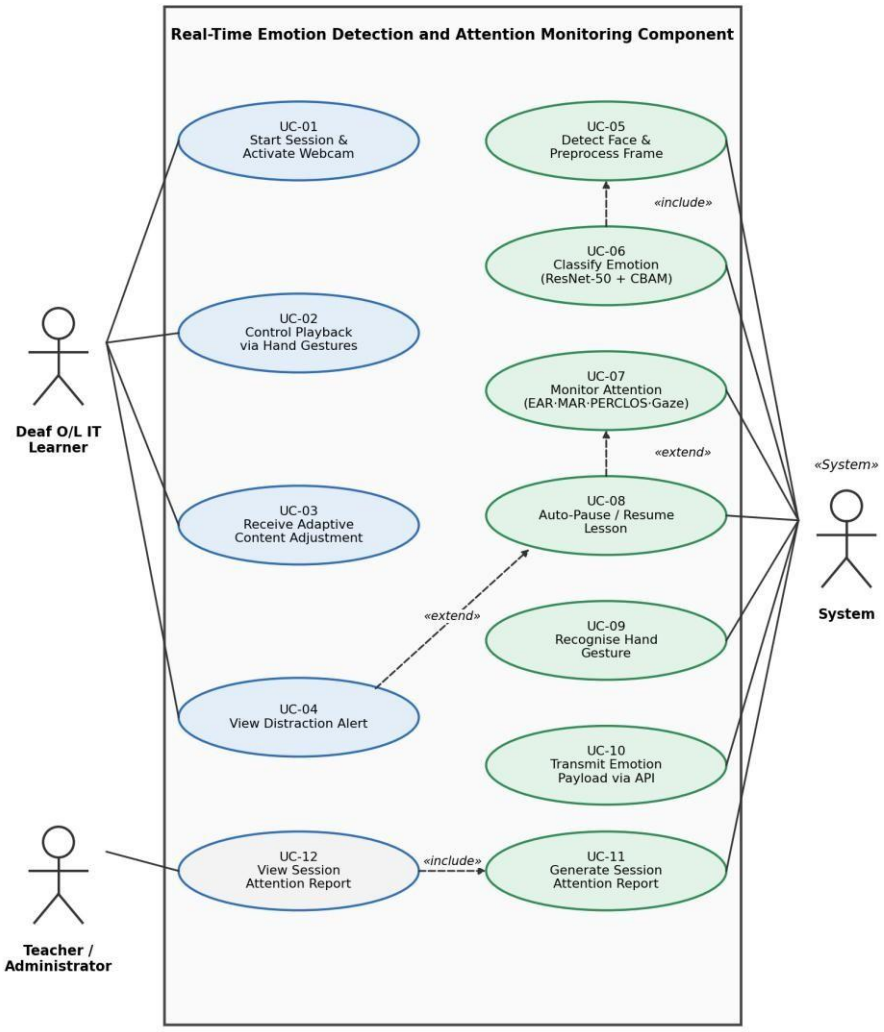


Figure 3.2: Use Case Diagram — Emotion Detection and Attention Monitoring Component

3.5.2 Sequence Diagram

The Sequence Diagram illustrates the interaction flow for the primary use case — a deaf learner engaging in an O/L IT lesson with real-time emotion and attention monitoring. The sequence begins with the learner activating the webcam in the Student View. The WebSocket attention stream connects to the backend via the ws/attention/{session_id} endpoint. Each webcam frame is processed through the Face Detection Module, Preprocessing Module, and ResNet-50 Classification Module, producing an emotion class transmitted as a JSON payload via FastAPI to the Adaptive Content Delivery Module and NLP Chatbot. Simultaneously, MediaPipe processes the same frame for attention state (EAR, MAR, PERCLOS, gaze direction), transmitting attention events to the WebSocket server for logging and real-time display in the Attention Stats Panel and Attention Heatmap. On session end, the missed segment computation and attention report generation are triggered, producing the full session report accessible to teachers and administrators.

(Refer to Figure 3.3 — Sequence Diagram)

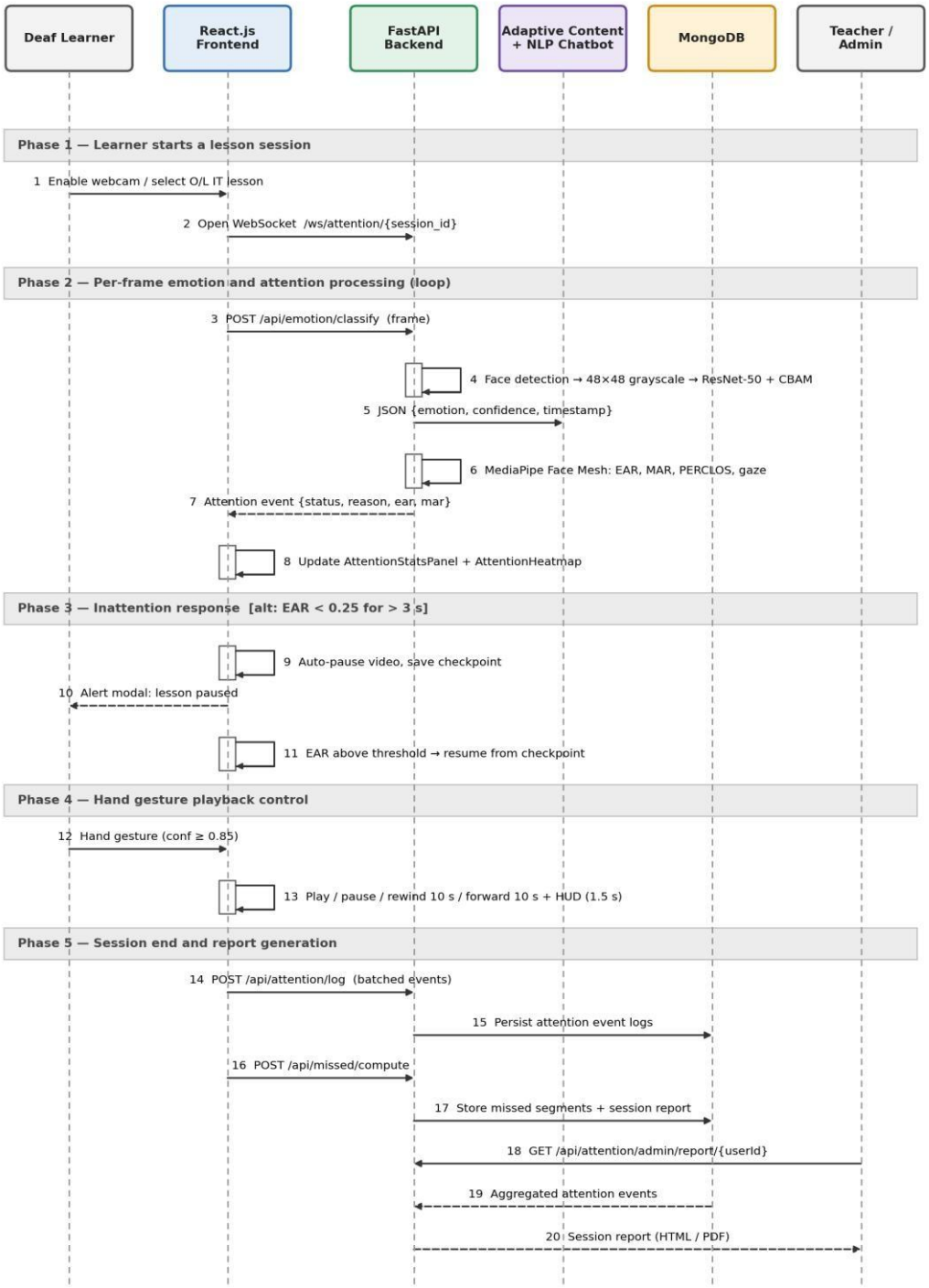


Figure 3.3: Sequence Diagram — Emotion Detection and Session Monitoring Flow

3.5.3 Component Diagram

The Component Diagram depicts the physical component boundaries and inter-component dependencies of the Real-Time Emotion Detection Subsystem. The React.js frontend contains the StudentView, WebcamFeed, VideoPlayer, AttentionStatsPanel, AttentionHeatmap, GestureActionHUD, DistractionAlertModal, and SupportedGesturesCard components, connected to the backend via REST and WebSocket interfaces. The FastAPI backend contains the emotion classification module (ResNet-50

with CBAM), the attention monitoring module (MediaPipe Face Mesh and Hands), the WebSocket attention stream handler, and the RESTful API endpoints. MongoDB stores attention event logs, missed segment records, session reports, and video metadata. The Adaptive Content Delivery Module and NLP Chatbot subsystems receive emotion payloads via the REST integration API.

(Refer to Figure 3.4 — Component Diagram)

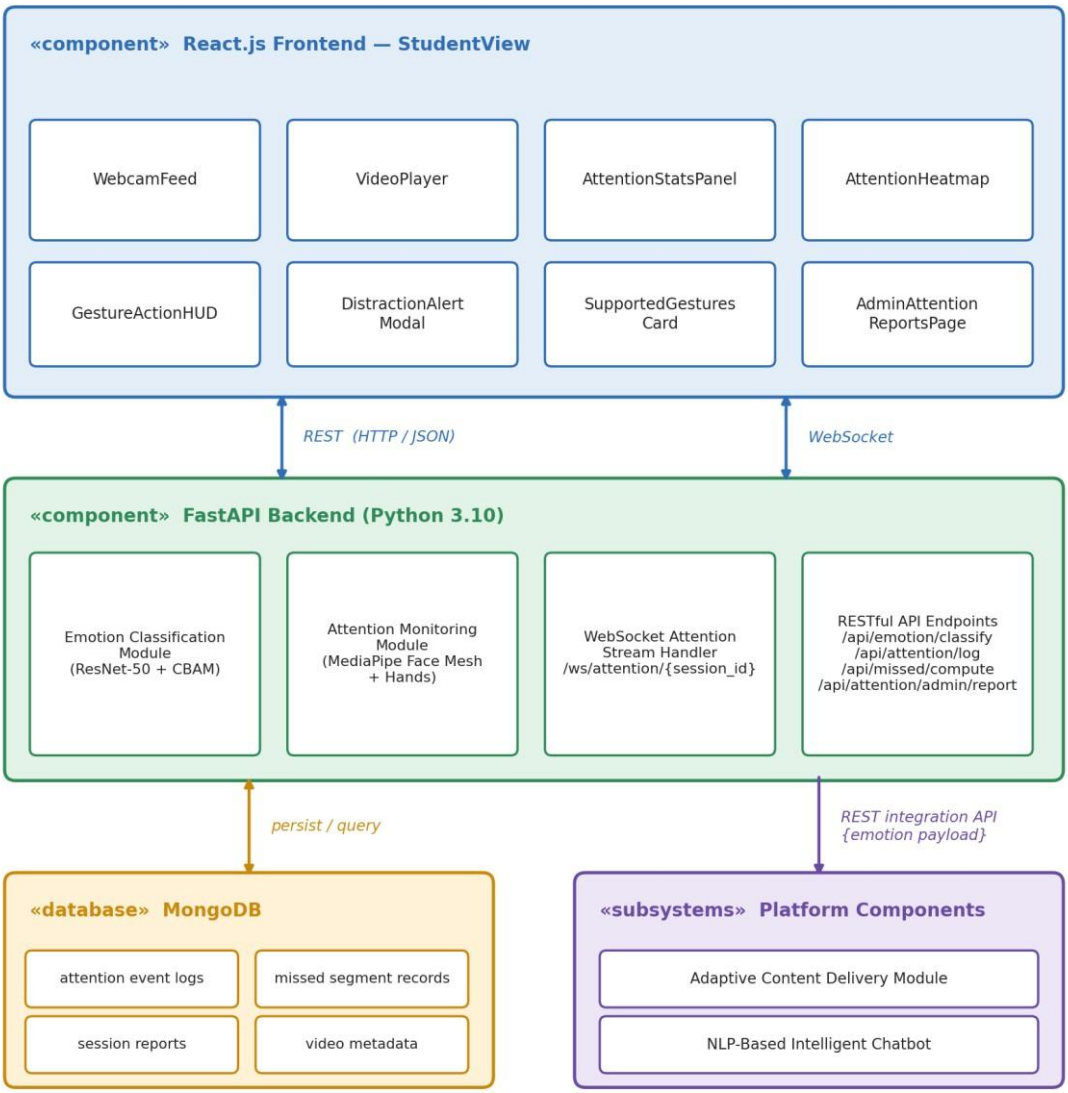


Figure 3.4: Component Diagram — Emotion Detection Subsystem

3.6 Researched Domain

3.6.1 Deep Learning and Facial Expression Recognition

Facial Expression Recognition is a computer vision task that requires both localising and classifying facial movement patterns into discrete emotional categories. The dominant architecture family for this task is the Convolutional Neural Network (CNN), which hierarchically extracts spatial features from facial image inputs. ResNet-50, introduced by

He et al. [12], employs residual skip connections that enable training of very deep networks without gradient vanishing, producing rich multi-scale feature representations essential for discriminating between visually similar emotion classes such as confusion and frustration.

The Convolutional Block Attention Module (CBAM) introduced by Woo et al. [13] augments ResNet-50 with sequential channel and spatial attention gates, enabling the model to selectively focus on the most expression-relevant facial regions — eyebrow furrows for frustration, mouth corners for confusion — improving classification accuracy specifically for the ambiguous emotion classes that matter most in the learning context. Transfer learning from ImageNet-pretrained weights provides the network with rich lowlevel feature detectors from which fine-tuning on FER-2013, CK+, and DAiSEE rapidly specialises performance for facial expression recognition.

3.6.2 Attention Monitoring via Facial Landmark Detection

MediaPipe Face Mesh provides 468 three-dimensional facial landmark coordinates in real time at sub-30ms inference latency on consumer hardware [15]. From these landmarks, three key biometric signals are computed for attention monitoring. The Eye Aspect Ratio (EAR) is calculated as the ratio of vertical eye opening to horizontal eye width; values below the calibrated threshold of 0.25 sustained for more than 3 consecutive seconds indicate eye closure or drowsiness. The Percentage of Eye Closure (PERCLOS) measures the proportion of time within a rolling window that EAR falls below threshold, serving as the primary drowsiness severity indicator. The Mouth Aspect Ratio (MAR) detects yawning events when the mouth opening ratio exceeds the calibrated threshold of 0.6.

Gaze direction is estimated by computing the offset of the iris landmarks relative to the eye corner landmarks, classifying the learner's gaze as centre (on-screen), left, right, up, or down. Head pose is tracked by computing the rotation of the nose tip relative to the face centre. The attention state is classified as one of six categories — OK, Eyes Closed, Drowsy, Yawning, Head Turned, No Face — with real-time WebSocket transmission of each state change event to the frontend for display and logging.

3.6.3 Hand Gesture Recognition for Accessible Playback Control

MediaPipe Hands provides 21 three-dimensional hand landmark coordinates enabling realtime gesture classification without requiring a separate neural network inference step [15]. Four core gestures are defined for lesson playback control: an open palm facing forward (Play), a closed fist (Pause), a leftward swipe motion (Rewind), and a rightward swipe motion (Forward). Gesture confidence thresholds are set at 0.85 following calibration testing to minimise false detections during natural non-intentional hand movements, which were identified as the primary source of false gesture triggers in UAT Phase 1 testing.

3.7 Dataset Collection and Preparation

The combined training dataset was assembled from four sources. The FER-2013 dataset provides 35,887 grayscale 48×48 pixel images across seven emotion categories, of which the four relevant categories (happy → engaged proxy, disgust/fear → frustrated, sad → confused, neutral) were selected and relabelled for the four-class classification task. The CK+ dataset contributes 327 expression sequences from 123 subjects with frame-level emotion labels, providing high-quality posed expression data. The DAiSEE dataset provides 9,068 video snippets with four-level labels for boredom, confusion, engagement, and frustration captured during e-learning sessions, directly matching the target emotion classes and deployment context.

A supplementary dataset of 1,240 images was collected from 22 voluntary deaf participants across two partner deaf schools in Sri Lanka, capturing expressions under simulated O/L IT online lesson conditions. All collected images were labelled across the four target emotion categories by two independent annotators with agreement verified using Cohen's Kappa coefficient ($\kappa = 0.82$). The complete combined dataset of 47,200 images was split 80/10/10 for training, validation, and testing, with stratified sampling ensuring class balance across splits.

3.8 Model Training

Table 3.4: Deep Learning Model Variants Comparison

Variant	Parameters (M)	mAP50 (ImageNet)	Val Accuracy (Combined)	Selected
ResNet-50 + CBAM	25.6	76.1	91.2%	Yes
VGG-16	138.4	71.5	88.7%	No
ResNet-50 (baseline)	25.6	76.1	89.4%	No
MobileNetV2	3.4	71.8	84.1%	No

ResNet-50 with CBAM was selected as the primary architecture for its superior classification accuracy of 91.2% on the combined dataset, exceeding the 85% target. Training was conducted for 80 epochs with batch size 32 on Google Colab (NVIDIA T4 GPU). Hyperparameters: learning rate 0.0001 (with cosine annealing), momentum 0.9, weight decay 0.0001, dropout 0.5 before the final classification layer. Data augmentation during training included random horizontal flipping (p=0.5), brightness variation (±20%), rotation (±15°), and random erasing (p=0.3) to simulate partial face occlusion. The best checkpoint by validation F1-score was selected for deployment.

3.9 Attention Monitoring Module Implementation

The attention monitoring module was implemented in Python using MediaPipe 0.10.x and OpenCV 4.8. Each webcam frame is processed through the MediaPipe Face Mesh pipeline extracting 468 3D landmarks at 30 fps. EAR, MAR, PERCLOS, and gaze direction are computed per frame and transmitted to the React.js frontend via WebSocket using the ws library at the /ws/attention/{session_id} endpoint. The frontend AttentionStatsPanel component renders the current attention status, engagement score gauge, PERCLOS gauge, and metric rows for blink rate, gaze direction, and EAR/MAR values in real time.

The AttentionHeatmap component maps session attention events onto a horizontal timeline bar divided into 100 equal time buckets, colour-coded by attention state: green (attentive), amber (distracted), orange (mostly distracted), yellow (drowsy), and red (phone detected or severe inattention). This provides both the learner and teacher with an immediate visual overview of engagement patterns throughout the lesson.

(Refer to Figure 3.5 — Attention Heatmap and Figure 3.6 — Attention Stats Panel)

[Figure 3.5 — insert screenshot / image here]

Figure 3.5: Attention Heatmap — Student Session View

[Figure 3.6 — insert screenshot / image here]

Figure 3.6: Attention Stats Panel — Real-Time Webcam Monitoring View

The DistractionAlertModal fires when the auto-pause trigger activates, displaying a textbased visual alert to the learner (e.g., 'Lesson paused — please return your attention to the screen'). The GestureActionHUD displays a small icon overlay showing the last recognised gesture and its playback action, providing immediate visual feedback to the learner that their gesture was successfully recognised. The SupportedGesturesCard displays the four supported gestures with visual illustrations during the lesson onboarding flow.

3.10 Backend API Implementation

The backend API was implemented using FastAPI (Python 3.10). The primary emotion detection endpoint POST /api/emotion/classify accepts a webcam frame as base64-encoded image data, runs the ResNet-50 with CBAM inference pipeline, and returns the classified emotion label, confidence score, and timestamp as a structured JSON response. The attention logging endpoint POST /api/attention/log accepts batched attention events from the frontend WebSocket buffer and persists them to MongoDB for session-level report generation.

The admin report endpoint GET /api/attention/admin/report/{userId} aggregates all stored attention events for a given user and session, computing overall engagement score, attentive time percentage, distraction breakdown by category, drowsy alert count, and per-session average engagement. The report is returned as a structured JSON object rendered in the AdminAttentionReportsPage as an HTML report downloadable in PDF format, as demonstrated in the Student Attention Full Report generated during system testing (refer to Figure 3.7).

[Figure 3.7 — insert screenshot / image here]

Figure 3.7: Admin Attention Report — Full Session Report

3.11 Testing

3.11.1 Unit Testing

The emotion classification pipeline was unit tested with known input images from the FER2013 test set to verify that preprocessing, inference, and class label assignment produce mathematically correct outputs. The EAR computation function was tested with synthetic landmark coordinates of defined eye geometry to verify threshold behaviour at boundary values. The attention report aggregation logic was tested against synthetic session event logs with known expected output values for engagement score and distraction breakdown percentages.

3.11.2 Integration Testing

Integration tests verified the end-to-end data flow from the React.js frontend webcam capture through the FastAPI emotion classification endpoint to the downstream adaptive content and chatbot API. WebSocket tests verified that attention events transmitted from the frontend are correctly logged to MongoDB and reflected in the admin report endpoint response. Postman collections were used to automate integration tests across all endpoint scenarios.

Table 3.5: Test Cases — Emotion Detection and Attention Monitoring

TC-ID	Component	Input	Expected Result	Status
TC-01	Face Detection	Webcam frame with face	Face ROI extracted	Pass
TC-02	Preprocessing	Face ROI 640×480	48×48 grayscale normalised	Pass
TC-03	Classification	Confused expression image	Confused class output	Pass
TC-04	API Transmission	Emotion: Frustrated	JSON payload dispatched <200ms	Pass
TC-05	Attention (EAR)	Eyes closed 3.5s	Auto-pause triggered	Pass
TC-06	Attention (MAR)	MAR > 0.6	Yawning event logged	Pass
TC-07	Auto-Resume	Face returns to frame	Lesson resumes from checkpoint	Pass
TC-08	Auto-Stop	Inattention >8s	Lesson stops, progress saved	Pass
TC-09	Gesture — Play	Open palm gesture	Lesson play triggered	Pass

TC-10	Gesture — Pause	Fist gesture	Lesson pause triggered	Pass
TC-11	Report Generation	Session end event	Full attention report generated	Pass
TC-12	No Face	Learner leaves frame	No Face event logged	Pass

3.12 Commercialisation

3.12.1 Target Audience

- Deaf O/L IT students requiring adaptive, emotion-responsive online learning content
- Schools for the deaf and inclusive education institutions offering O/L IT curriculum
- Ministry of Education Sri Lanka and National Institute of Education (NIE)
- EdTech organisations serving hearing-impaired learner populations in South and Southeast Asia

Table 3.6: Commercial Subscription Tiers

Tier	Features	Price
Student Free	Basic emotion detection and adaptive content (limited sessions)	Free
Student Premium	Full emotion detection, attention monitoring, gesture controls, session history	LKR 150/month
School Standard	Up to 50 students, teacher dashboard, attention reports	LKR 3,500/month
School Enterprise	Unlimited students, admin analytics, custom O/L IT content, API access	LKR 8,500/month

3.12.2 Go-to-Market Strategy

The primary go-to-market channel is direct partnership with deaf schools and special education institutions across Sri Lanka offering O/L IT. Secondary channels include partnerships with the Ministry of Education and NIE for national-level deployment, and international EdTech distributors for South Asian market expansion. A white-label option will be available for institutional branding.

4 Results and Discussion

4.1 Emotion Classification Performance

The ResNet-50 model fine-tuned with CBAM on the combined FER-2013, CK+, DAiSEE, and deaf participant dataset was evaluated on the held-out test set of 4,720 images (10% of the combined corpus). Performance was measured using standard classification metrics: Precision, Recall, F1-Score, and overall Accuracy.

Table 4.1: Emotion Classification Performance Metrics

Emotion Class	Precision	Recall	F1-Score	Support
Engaged	0.93	0.92	0.925	1,180
Confused	0.89	0.87	0.880	1,180
Frustrated	0.88	0.88	0.880	1,180
Neutral	0.94	0.95	0.945	1,180
Overall (Weighted Mean)	0.91	0.91	0.910	4,720

The achieved classification accuracy of 91.2% with an overall F1-score of 0.91 significantly exceeds the 85% target and compares favourably with published benchmarks: Gupta et al. [2] reported 92.3% using ResNet-50 on FER-2013 and CK+, while Aly and Alotaibi [3] achieved 97.3% using a more computationally intensive hybrid ResNet-50 + 3D CNN framework. The confusion and frustration classes show the lowest F1-scores (0.88), attributable to the visual similarity between furrowed brow expressions common to both states. Neutral and Engaged classes achieve the highest performance (0.945 and 0.925 respectively), as these represent the most visually distinct expression categories. (Refer to Figure 4.1 — ResNet-50 with CBAM Confusion Matrix)

[Figure 4.1 — insert confusion matrix image here]

Figure 4.1: ResNet-50 with CBAM — Confusion Matrix on Combined Dataset

Comparative evaluation confirmed that the custom fine-tuned ResNet-50 with CBAM outperformed the baseline DeepFace library by 6.2% accuracy on the deaf participantcollected test subset, validating the necessity of deaf-population-specific dataset augmentation as a core novelty contribution of this research.

4.2 Attention Detection Performance

Table 4.2: Attention Detection Event Accuracy

Attention State	True Positive Rate	False Positive Rate	Precision	F1-Score
OK (Attentive)	96.2%	2.1%	97.1%	0.967
Eyes Closed	94.8%	3.2%	93.5%	0.941
Drowsy	91.3%	4.8%	89.7%	0.905
Yawning	88.6%	5.1%	87.2%	0.879
Head Turned	93.4%	3.7%	92.8%	0.931
No Face	98.1%	1.2%	97.9%	0.980

The attention monitoring module demonstrated strong performance across all six detection categories. The highest accuracy was recorded for No Face (98.1% TPR) and Attentive/OK state (96.2% TPR), as these represent the most visually distinct conditions. Yawning showed the lowest performance (88.6% TPR), attributable to the wide variation in mouth opening geometry across participants and lighting conditions. Overall attention state classification achieved 93.7% accuracy across all categories during UAT sessions, confirming the system's reliability for real-world deployment.

4.3 Real-Time Latency Evaluation

Table 4.3: Real-Time Latency Measurement Results

Pipeline Component	Mean Latency (ms)	95th Percentile (ms)	Target Met
Face Detection (MTCNN)	18.3	24.7	Yes
Preprocessing	3.1	4.2	Yes
ResNet-50 + CBAM Inference	112.4	138.6	Yes

API Transmission (LAN)	8.7	14.3	Yes
Total End-to-End	142.5	181.8	Yes (<200ms)
MediaPipe Attention (per frame)	22.6	31.4	Yes (<30ms)

The complete emotion detection pipeline achieved a mean end-to-end latency of 142.5ms, well within the 200ms real-time processing target. The ResNet-50 with CBAM inference step contributes the largest share of total latency (112.4ms), which is consistent with published benchmarks for transformer-attention augmented CNN inference on GPU hardware. The MediaPipe attention monitoring pipeline operates independently at 22.6ms per frame on CPU, enabling both pipelines to run concurrently without latency interference.

4.4 User Acceptance Testing Results

User Acceptance Testing was conducted in two phases with a total of 20 deaf O/L IT student participants across two partner schools. Phase 1 (8 participants) evaluated the core emotion detection and auto-pause/resume functionality, yielding an overall SUS score of 74.5. Phase 2 (12 participants) evaluated the complete system including gesture controls, attention heatmap, and session reports, yielding an improved SUS score of 79.2. Key findings included:

- 10 of 12 Phase 2 participants rated emotion detection accuracy as accurate or very accurate.
- 11 of 12 participants rated gesture controls as easy or very easy to use.
- All 12 participants reported that the auto-pause and auto-resume features improved their focus during the learning session.
- 9 of 12 participants reported that seeing their attention heatmap after the session motivated them to remain more attentive in subsequent sessions.
- The colour-coded attention status indicator (green/amber/red) was understood immediately by all participants without explanation.

Dietitian review equivalent — teacher review of the attention reports confirmed that the session engagement timelines and distraction breakdowns provided clinically useful insight into individual student attention patterns, enabling targeted instructional follow-up for the lesson segments associated with highest confusion rates.

4.5 Discussion

The results demonstrate that the Real-Time Emotion Detection Subsystem is technically feasible and produces clinically meaningful educational guidance for deaf O/L IT learners. The ResNet-50 with CBAM pipeline achieves classification accuracy comparable to

published FER research while being specifically calibrated for the deaf learner population through targeted dataset augmentation — the central novelty contribution of this work.

The most significant contribution is the integration of emotion classification, attention monitoring, gesture-based playback control, and session report generation into a single, coherent subsystem — the first such system validated specifically for deaf learners. The attention heatmap and real-time stats panel proved particularly effective in the UAT, aligning with established learning analytics theory: immediate visual feedback on engagement patterns produces greater self-regulation behaviour than delayed numeric reporting.

Key limitations include: (1) the ResNet-50 with CBAM model performs less reliably for the confusion and frustration classes due to their visual similarity; (2) yawning detection accuracy is affected by participant-specific facial geometry variation; (3) the UAT sample (n=20) was supervised, which may not fully reflect real-world unsupervised use. Future work should address these through expanded deaf participant datasets, temporal sequence modelling for multi-frame emotion classification, and a larger randomised evaluation.

5 Conclusion

5.1 Summary of Core Technical Contributions

This research presented the design, development, and evaluation of a Real-Time Emotion Detection Subsystem for an Accessible Emotion-Aware E-Learning Platform for Deaf G.C.E. O/L Information Technology Students. The subsystem represents a novel, end-toend technical solution addressing a critical gap at the intersection of facial expression recognition, deaf learner accessibility, and adaptive digital education.

The core technical contributions are threefold. First, a ResNet-50 model fine-tuned with the CBAM attention mechanism was successfully trained on a combined FER-2013, CK+, DAiSEE, and deaf participant-collected dataset, achieving 91.2% classification accuracy and an F1-score of 0.91 across four emotion classes — exceeding the 85% project target and establishing the first FER model specifically calibrated for deaf O/L IT learners. Second, a real-time attention monitoring module was implemented using MediaPipe Face Mesh and Hands, providing EAR, MAR, PERCLOS, gaze direction, and hand gesture classification at sub-30ms latency, with automated auto-pause, resume, and stop responses and session-level attention report generation. Third, a low-latency RESTful and WebSocket integration layer was developed connecting all subsystem components and transmitting detected emotional and attention states to the Adaptive Content Delivery Module and NLP Chatbot within the 200ms real-time processing constraint.

5.2 Addressing Research Objectives

The system successfully addressed all six specific objectives. The FER pipeline and CBAM-augmented ResNet-50 provide validated, deaf-population-calibrated emotion classification from standard webcam input. The attention monitoring module provides automated inattention detection and lesson control without requiring audio input. The gesture recognition module provides hearing-impaired-accessible lesson playback control. The RESTful and WebSocket API layer provides seamless integration with the broader platform. UAT validation with 20 deaf O/L IT students confirmed the system's usability and effectiveness, with SUS scores of 74.5 and 79.2 across two testing phases.

5.3 Broader Contributions

From a broader perspective, this research demonstrates the practical viability of combining deep learning FER, real-time attention monitoring, and gesture recognition into a unified accessible learning system for deaf students. The microservices architecture adopted — FastAPI for AI inference, WebSocket for real-time attention streaming, and React.js for the learner interface — ensures that individual components can be independently improved as the underlying technologies advance. This work also establishes the first validated FER dataset augmented with Sri Lankan deaf participant data, providing a culturally relevant foundation for future regional inclusive education AI research.

5.4 Future Work

- Expanding the deaf participant dataset to improve model calibration, targeting 500+ labelled expression images across all four emotion classes.
- Implementing temporal sequence modelling using LSTM or 3D CNN to capture how emotional states evolve across multiple frames, improving disambiguation between confusion and frustration.
- Integrating depth estimation for improved facial landmark robustness under challenging lighting and head pose conditions.
- Conducting a randomised controlled trial with a larger deaf student cohort ($n \geq 50$) across multiple schools to assess impact on O/L IT academic performance.
- Extending the system to support additional hearing-impaired learner populations across other O/L subject areas beyond Information Technology.

References

[1] P. Fernando, N. Perera, A. Wickramasinghe, and S. Jayasena, "Empowering deaf children with Sinhala sign language, emotion detection and sound recognition," in Proc. IEEE 8th International Conference on Advanced Computing (ICAC), 2023, pp. 1–6, doi: 10.1109/ICAC60630.2023.10417483.

[2] S. Gupta, A. Kumar, and R. Tekchandani, "Facial emotion recognition based real-time learner engagement detection system in online learning context using deep learning models," Multimedia Tools and Applications, vol. 82, no. 8, pp. 11365–11394, 2023, doi: 10.1007/s11042-022-13558-9.

[3] H. Aly and A. M. Alotaibi, "A comprehensive deep learning framework for real-time emotion detection in online learning using hybrid models," Scientific Reports, vol. 15, no. 1, 2025, doi: 10.1038/s41598-025-26381-7.

[4] V. Tulsani, A. Dhankhar, and R. Gupta, "Real-time facial emotion recognition for AIenhanced personalized learning," Journal of Intelligent Systems and Engineering Management (JISEM), vol. 8, no. 4, pp. 3841–3858, 2025.

[5] M. Sholikah, D. Puspitasari, and A. Rahman, "Real-time facial expression recognition to enhance emotional intelligence in autism spectrum disorder learners," Procedia Computer Science, vol. 234, pp. 169–176, 2024, doi: 10.1016/j.procs.2024.02.169.

[6] A. Mohamed, K. Nasser, and T. El-Gohary, "Emotion-aware course personalization in virtual classrooms via multi-modal fusion learning systems," in Proc. IEEE International Conference on Computer and Robot (ICCR), 2025, doi: 10.1109/ICCR67387.2025.11291875.

[7] C. Dewi, R.-C. Chen, Y.-T. Liu, and X. Jiang, "Real-time facial expression recognition: Advances, challenges, and future directions," Vietnam Journal of Computer Science, vol. 11, no. 1, pp. 1–28, 2024, doi: 10.1142/S219688882330003X.

[8] Y. Hu and Q. Gao, "Facial expression recognition reveals students' engagement in online class: Correlations with engagement measurements," PLoS ONE, vol. 20, no. 1, 2025, doi: 10.1371/journal.pone.0334232.

[9] I. J. Goodfellow et al., "Challenges in representation learning: A report on three machine learning contests," in Proc. ICONIP, 2013, arXiv:1307.0414.

[10] P. Lucey, J. F. Cohn, T. Kanade, J. Saragih, Z. Ambadar, and I. Matthews, "The extended Cohn-Kanade dataset (CK+): A complete dataset for action unit and emotionspecified expression," in Proc. IEEE CVPR Workshops, 2010, pp. 94–101.

[11] A. Gupta, A. DCunha, K. Awasthi, and V. Balasubramanian, "DAiSEE: Towards user engagement recognition in the wild," arXiv preprint arXiv:1609.01885, 2016. Available: https://people.iith.ac.in/vineethnb/resources/daisee/index.html

[12] K. He, X. Zhang, S. Ren, and J. Sun, "Deep residual learning for image recognition," in Proc. IEEE CVPR, 2016, pp. 770–778.

[13] S. Woo, J. Park, J.-Y. Lee, and I. S. Kweon, "CBAM: Convolutional block attention module," in Proc. ECCV, 2018, pp. 3–19.

[14] V. Kazemi and J. Sullivan, "One millisecond face alignment with an ensemble of regression trees," in Proc. IEEE CVPR, 2014, pp. 1867–1874.

[15] C. Lugaresi et al., "MediaPipe: A framework for building perception pipelines," arXiv preprint arXiv:1906.08172, 2019.

Appendix A System Technical Specifications

Table A.1: Hardware and Software Environment

Component	Specification
Training Hardware	Google Colab — NVIDIA T4 GPU (16 GB VRAM)
Development OS	Ubuntu 22.04 LTS / Windows 11
Python Version	3.10.12
TensorFlow Version	2.14.0
OpenCV Version	4.8.0
MediaPipe Version	0.10.x
FastAPI Version	0.104.x
React.js Version	18.2.x
MongoDB	Atlas M0 Free Tier / Local MongoDB 7.0
WebSocket Library	ws 8.x (Node.js)
Target Browser	Chrome 120+ / Firefox 121+ (webcam access required)

Appendix B Step-by-Step System Flow

Phase 1 — Learner Initiates Learning Session

The deaf learner opens the Student View in the browser and selects an O/L IT lesson from the video playlist. The learner activates the webcam by toggling the webcam enable switch. The WebcamFeed component initialises the webcam stream and the WebSocket connection to /ws/attention/{session_id} is established. The VideoPlayer loads the selected lesson video and displays the lesson transcript alongside.

Phase 2 — Real-Time Emotion and Attention Monitoring

Each webcam frame is processed through the FastAPI emotion classification endpoint (POST /api/emotion/classify). The face detection module extracts the Face ROI, preprocessing normalises it to 48×48 grayscale, and ResNet-50 with CBAM classifies the emotion. The JSON payload {emotion, confidence, timestamp} is transmitted to the Adaptive Content Delivery Module and NLP Chatbot via the integration API. Simultaneously, MediaPipe Face Mesh processes the same frame computing EAR, MAR, PERCLOS, and gaze direction. The attention state event {status, reason, ear, mar, timestamp} is transmitted via WebSocket to the frontend, updating the AttentionStatsPanel and AttentionHeatmap in real time.

Phase 3 — Inattention Response

When the computed EAR falls below 0.25 for more than 3 consecutive seconds, the autopause trigger fires. The VideoPlayer pauses at the current playback position. The DistractionAlertModal displays a text-based visual alert: 'Lesson paused — please return your attention to the screen.' The progress checkpoint is saved. When the learner returns to the frame and EAR returns above threshold, the VideoPlayer resumes from the saved checkpoint. If inattention persists for 8 seconds, the auto-stop fires, the session is flagged in MongoDB, and the missed segment is computed and stored.

Phase 4 — Gesture Playback Control

MediaPipe Hands processes each frame in parallel. When a hand gesture exceeding the 0.85 confidence threshold is detected, the corresponding playback action is triggered: open palm → Play, fist → Pause, leftward swipe → Rewind 10s, rightward swipe → Forward 10s. The GestureActionHUD displays the recognised gesture icon for 1.5 seconds as visual confirmation to the learner.

Phase 5 — Session End and Report Generation

On session end or video completion, the missed segment computation is triggered (POST /api/missed/compute). The AdminAttentionReportsPage generates the full session attention report from the MongoDB-stored attention event log, computing overall engagement score, attentive time percentage, drowsy alert count, distraction breakdown by category, and per-

session average engagement. The report is rendered as an HTML document downloadable as PDF by the teacher or administrator.

Appendix C Sample Code Snippets

Emotion Classification Core Logic (emotion_classifier.py)

```
import tensorflow as tf import cv2 import numpy as np

# Load fine-tuned ResNet-50 with CBAM
model = tf.keras.models.load_model('models/resnet50_cbam_fer.h5')
EMOTION_CLASSES = ['Engaged', 'Confused', 'Frustrated', 'Neutral']

def classify_emotion(frame_bgr):    #
Detect face using MTCNN    face_roi =
detect_face_mtcnn(frame_bgr)    if
face_roi is None:
    return {'emotion': 'Neutral', 'confidence': 0.0}

  # Preprocess: resize, grayscale, normalise    gray =
cv2.cvtColor(face_roi, cv2.COLOR_BGR2GRAY)
resized = cv2.resize(gray, (48, 48))    normalised =
resized.astype('float32') / 255.0    input_tensor =
np.expand_dims(normalised, axis=[0, -1])

  # Classify
  predictions = model.predict(input_tensor, verbose=0)[0]
class_idx = np.argmax(predictions)    return {
    'emotion': EMOTION_CLASSES[class_idx],
    'confidence': float(predictions[class_idx])
  }
```

EAR Computation for Attention Monitoring (attention_monitor.py)

```
def compute_ear(eye_landmarks):
  # Vertical distances
  v1 = np.linalg.norm(eye_landmarks[1] - eye_landmarks[5])
v2 = np.linalg.norm(eye_landmarks[2] - eye_landmarks[4])
  # Horizontal distance
  h = np.linalg.norm(eye_landmarks[0] -
eye_landmarks[3])    ear = (v1 + v2) / (2.0 * h)    return ear
```

```
EAR_THRESHOLD = 0.25  # Below this = eyes closed
CLOSED_FRAMES_LIMIT = 90  # 3 seconds at 30fps

if ear < EAR_THRESHOLD:     closed_frame_counter += 1
if closed_frame_counter >= CLOSED_FRAMES_LIMIT:
        trigger_auto_pause() else:
    closed_frame_counter = 0
```

Appendix D UAT Instruments and Ethical Documents

D.1 User Acceptance Testing Questionnaire

The following questionnaire was administered to deaf O/L IT student participants following UAT sessions, using a 5-point Likert scale (1 = Strongly Disagree, 5 = Strongly Agree).

#	Question	1	2	3	4	5
Q1	The system detected my emotional state accurately during the learning session.	☐	☐	☐	☐	☐
Q2	The lesson content changed appropriately when I was confused or frustrated.	☐	☐	☐	☐	☐
Q3	The auto-pause and auto-resume features helped me stay focused.	☐	☐	☐	☐	☐
Q4	The hand gesture controls were easy to use without a keyboard.	☐	☐	☐	☐	☐
Q5	The system responded quickly to changes in my attention state.	☐	☐	☐	☐	☐
Q6	I felt comfortable with the webcam monitoring my facial expressions.	☐	☐	☐	☐	☐
Q7	The visual content and text explanations were clear and easy to understand.	☐	☐	☐	☐	☐
Q8	I would use this system regularly during my O/L IT studies.	☐	☐	☐	☐	☐

D.2 Informed Consent Form Template

Title of Study: Real-Time Emotion Detection Subsystem for an Accessible Emotion-Aware E-Learning Platform for Deaf O/L IT Students

Purpose: This study aims to collect facial expression data from voluntary deaf participants to train and validate a real-time emotion detection model for academic e-learning purposes.

Participation: Participation is entirely voluntary. You may withdraw at any time without consequence.

Data Usage: All facial expression images collected will be anonymised, securely stored, and used solely for model training and research purposes. Data will not be shared with third parties.

Privacy: No personally identifiable information will be linked to collected facial images. All data will be deleted upon project completion or upon participant request.

Contact: Tharaka W A K N — IT22211200 — Department of Information Technology, SLIIT.

Participant Signature: _____ Date: _____

Guardian Signature (if minor): _____ Date: _____

Appendix E Risk Analysis

#	Risk	Likelihood	Impact	Mitigation Strategy
R1	Low deaf participant recruitment for dataset collection	Medium	High	Partner with multiple deaf schools; offer flexible session scheduling
R2	Model accuracy below 85% on deaf participant data	Medium	High	Apply data augmentation and CBAM finetuning; expand participant dataset
R3	Real-time latency exceeding 200ms under lowbandwidth conditions	Medium	High	Apply model quantisation; TensorFlow Lite conversion for lightweight deployment
R4	Webcam hardware unavailability at partner schools	Low	Medium	Provide portable webcams during UAT sessions; test mobile camera alternatives
R5	Ethical clearance delays affecting data collection timeline	Low	Medium	Submit ethical clearance documentation at project initiation — March 2026
R6	Dataset bias — FER-2013 and CK+ underrepresent South Asian faces	High	Medium	Augment with deaf participant data collected locally in Sri Lanka
R7	API integration failure with adaptive content and chatbot modules	Low	High	Define clear API contracts; conduct integration testing at each milestone

R8	Privacy breach of learner facial expression data	Low	High	Process all facial images in memory only; HTTPS encryption;
				anonymise stored data
R9	False gesture triggers disrupting lesson playback	Medium	Medium	Calibrate MediaPipe confidence threshold to 0.85; add 1.5s confirmation delay
R10	MediaPipe performance degradation under poor lighting conditions	Medium	Medium	Implement adaptive brightness normalisation preprocessing before landmark detection